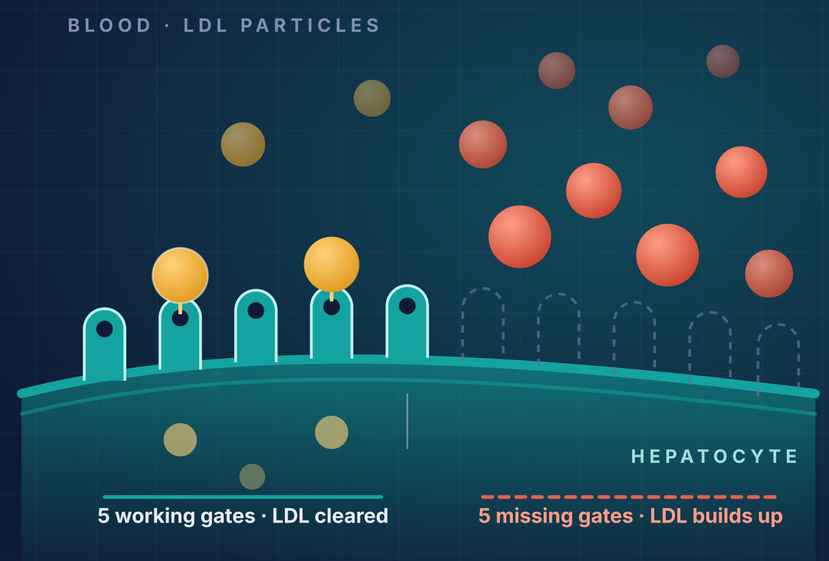
PHARMACOGENOMICS · CLINICAL PHARMACY

Locked *Out*

Genetics and statin response in familial hypercholesterolemia

Half the gates are missing. **How a gene keeps LDL out of the liver**, and how DNA shapes the drugs that let it back in.

Noor Al Ahmad • Clinical Pharmacy • Narrative review & proposed research question



1 in 311 — **Heterozygous FH** people worldwide carry one faulty allele

1 in 300k — **Homozygous FH** two faulty alleles, almost no receptors

≈3× — **Cardiovascular risk** in variant carriers versus non-carriers

>190 — **mg/dL LDL-C** (5 mmol/L) in adults; high from childhood

I The Gene

Why can't the liver let LDL in?

A 10-year-old with adult-level cholesterol

No symptoms. No warning. Only a variant inherited from one parent, and decades of silent exposure ahead. Risk tracks **LDL-C × years**: high LDL-C from birth means disease decades earlier.

INHERITED	The liver cannot clear LDL from the blood
LIFELONG	LDL-C very high from childhood, >190 mg/dL in adults
DANGEROUS	Premature atherosclerotic cardiovascular disease
DIAGNOSIS	LDL-C, family history, physical signs and a genetic test

FIG 1 Four genes, one broken gate



- **LDLR** 85–90% — The receptor itself is missing or defective
- **APOB** 5–10% — ApoB-100, the key on LDL, no longer fits
- **PCSK9** 1–3% — Gain of function: receptors destroyed too fast
- **LDLRAP1** <1% — Receptor not internalised (recessive)

Same end point: **fewer working LDL receptors → less clearance → higher LDL-C.**

FIG 2 Count the gates: one faulty allele or two

0 FAULTY ALLELES — No FH
- Full receptor activity
- Normal LDL clearance
- Statins work well

1 FAULTY ALLELE — Heterozygous
>190 mg/dL LDL-C
- About half the activity
- Statins usually work

2 FAULTY ALLELES — Homozygous
>400 mg/dL LDL-C
- Little or no activity
- Statins work poorly

Gate icons are schematic. Hu et al. 2020; Cuchel et al. 2023; Mach et al. 2020.

PCSK9: the demolition crew, switched off

Too much
Gain-of-function variants destroy receptors and **cause FH**
Abifadel 2003

Too little
Loss-of-function carriers
−88%
coronary disease, 28% lower LDL-C
Cohen 2006

Block it
Antibodies and siRNA
−50–60%
LDL-C; blunted in receptor-null homozygotes

II The Drug

Why does the same dose work differently?

FIG 3 Statins make the liver build more gates

1 · BLOCK — **HMG-CoA reductase** — The statin inhibits cholesterol synthesis

2 · SENSE — **SREBP-2 on** — Liver cholesterol falls; the sensor switches on

3 · BUILD — **More receptors** — More LDL receptors reach the cell surface

4 · CLEAR — **LDL-C falls** — By 30–50% or more as LDL leaves the blood

No working receptor allele → little statin effect. **Statin efficacy depends on the FH genotype.**

FIG 4 Same dose, different patients

Drug-handling genes (**SLCO1B1, ABCG2, CYP2C9**) set how much statin reaches blood and muscle; drug-target genes (**LDLR, APOB, PCSK9**) set how far LDL-C can fall. Adherence, interacting drugs, thyroid or kidney disease and diet add to both. **Handling genes** mainly drive **toxicity**; **target genes** mainly drive **efficacy**.

GENE	ROLE	STATIN MOST AFFECTED	EFFECT
SLCO1B1	Liver uptake door OATP1B1	Simvastatin	Higher myopathy risk
ABCG2	Efflux pump BCRP	Rosuvastatin	Higher drug exposure
CYP2C9	Metabolising enzyme	Fluvastatin	Slower clearance

CPIC 2022 turns each genotype into a statin- and dose-specific recommendation (Cooper-DeHoff et al.).

When the statin itself is locked out

SLCO1B1 c.521T>C — Less statin enters the liver, so more reaches muscle.

4.5× odds of myopathy **per C allele**

16.9× odds in **CC vs TT** homozygotes

CPIC 2022: choose another statin, or keep simvastatin below 20 mg/day. A risk genotype is **not** a reason to stop statins.

SEARCH Collaborative Group, NEJM 2008 (simvastatin 80 mg).

III The Patient

How does a genotype become a prescription?

FIG 5 Two ways out for LDL

Route 1 · Open more gates — NEEDS A WORKING RECEPTOR

Evolocumab, alirocumab anti-PCSK9 antibodies	50–60%
Inclisiran siRNA vs hepatic PCSK9	≈50%
Bempedoic acid ATP-citrate lyase inhibitor	15–25%
Ezetimibe NPC1L1 inhibitor	≈20%

Route 2 · Take another exit — RECEPTOR-INDEPENDENT

Evinacumab ANGPTL3 antibody	≈50%
Lomitapide MTP inhibitor	40–50%
Lipoprotein apheresis physical removal of LDL	procedure

0 20 40 60%

Approximate added LDL-C reduction in pivotal trials; route 2 values are from homozygous FH. Light segment = upper end of range. **When almost no gates remain, opening more of them cannot work.**

FIG 6 Four clues, one prescription

1 Disease genotype — Which gene, one allele or two, residual receptor activity

2 Pharmacogenotype — SLCO1B1, ABCG2 and CYP2C9 function

3 Phenotype — LDL-C, Lp(a), age, existing ASCVD

4 Response — LDL-C fall and tolerability at follow-up

℞ Together they decide **which drug, what dose, which add-on** and **how to monitor**.

Future directions: diet & bergamot

1. Genetic FH diagnosed
2. Lifestyle and heart-healthy diet
3. Statin therapy when indicated
4. Poor response or intolerance: individualised non-statin therapy
5. **Investigational adjunct: *Citrus bergamia* (bergamot) polyphenols**

DO NOT SAY — "A natural replacement for statins"

SAY INSTEAD — "A potential adjunct under investigation for additional LDL-C reduction"

Diet alone generally cannot overcome the genetic defect in FH.

PROPOSED RESEARCH QUESTION

Could genetically guided treatment combined with dietary and nutraceutical interventions improve LDL-C control in young patients with familial hypercholesterolemia?

Personalised lipid management in young patients with FH: from pharmacogenomics to dietary and nutraceutical interventions.

TAKE-HOME

1. FH genotype predicts **how well** statins can work.
2. SLCO1B1 · ABCG2 · CYP2C9 predict **how safely**.
3. Few gates left → choose a **receptor-independent** route.

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Noor Al Ahmad
Pharmacogenomics · Clinical Pharmacy

Half the gates are missing. DNA shows which key fits.